Linux Containers Lab Guide with Command Explanations and Screenshot Placeholders

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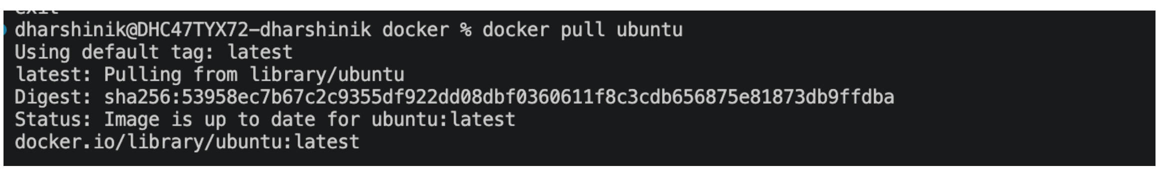
Step 1 - Pull Ubuntu Image

Command: docker pull ubuntu:24.04

Explanation: Downloads the Ubuntu Linux image from Docker Hub to your local machine. Images are templates used to create containers.

Expected Output: image pulled from docker hub

Screenshot Placeholder (Paste Screenshot Below):



```
dharshinik@DHC47TYX72-dharshinik docker % docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:53958ec7b67c2c9355df922dd08dbf0360611f8c3cdb656875e81873db9ffdba
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
```

Step 2 - Verify Image

Command: docker images

Explanation: Displays all images stored locally on your computer.

Expected Output: shows list of images

Screenshot Placeholder (Paste Screenshot Below):

```
dharshinik@DHC47TYX72-dharshinik docker % docker images

IMAGE                                 ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest                         28bd5fe8b56d       13.6MB         4.27MB    U
banking-api:1.0                       fdabb506d5ef        393MB         98.5MB    U
banking.api:1.0                       ae039ac8854d        393MB         98.5MB
compose-postgres-api-lab-api:latest   b880a79d9b55        387MB           97MB    U
postgres:16                           081f1bc7bd5e        663MB          165MB    U
postgres:latest                       29ee7bb30d80        671MB          167MB    U
ubuntu:latest                         53958ec7b67c        180MB         44.4MB    U
weatherapi:1.0                        0c50d191dbab        352MB         88.9MB
weatherapi:single                     a13326415044        1.3GB          320MB
```

Step 3 - Start Ubuntu Container

Command: docker run -it ubuntu:24.04 bash

Explanation: Creates and starts a container from the Ubuntu image and opens an interactive Bash shell.

Expected Output: Create and start a container. Attach it (execute within it in bash terminal)

Screenshot Placeholder (Paste Screenshot Below):

```
dharshinik@DHC47TYX72-dharshinik docker % docker run -it ubuntu bash
root@2cade0d6eaf7:/#
```

Step 4 - Check Current Directory

Command: pwd

Explanation: Prints the current working directory.

Expected Output: Print Working Directory

Screenshot Placeholder (Paste Screenshot Below):

```
root@2cade0d6eaf7:/# pwd
/
```

/ refers to root directory

Step 5 - List Files

Command: ls -la

Explanation: Displays all files and folders including hidden files.

Expected Output: list all files (including hidden) in long format (with permissions too)

Screenshot Placeholder (Paste Screenshot Below):

```
root@2cade0d6eaf7:/# ls -la
total 60
drwxr-xr-x   1 root root 4096 Jun 22 05:27 .
drwxr-xr-x   1 root root 4096 Jun 22 05:27 ..
-rwxr-xr-x   1 root root    0 Jun 22 05:27 .dockerenv
drwxr-xr-x   2 root root 4096 Jun 10 03:33 .rock
lrwxrwxrwx   1 root root    7 Apr 20 08:46 bin -> usr/bin
drwxr-xr-x   2 root root 4096 Apr 20 08:46 boot
drwxr-xr-x   5 root root  360 Jun 22 05:27 dev
drwxr-xr-x   1 root root 4096 Jun 22 05:27 etc
drwxr-xr-x   3 root root 4096 Jun 10 03:32 home
lrwxrwxrwx   1 root root    7 Apr 20 08:46 lib -> usr/lib
drwxr-xr-x   2 root root 4096 Jun 10 03:31 media
drwxr-xr-x   2 root root 4096 Jun 10 03:31 mnt
drwxr-xr-x   2 root root 4096 Jun 10 03:31 opt
dr-xr-xr-x 243 root root    0 Jun 22 05:27 proc
drwx------   2 root root 4096 Jun 10 03:32 root
drwxr-xr-x   4 root root 4096 Jun 10 03:32 run
```

Step 6 - View Linux Version

Command: cat /etc/os-release

Explanation: Displays operating system information.

Expected Output: To view the Linux OS versions

Screenshot Placeholder (Paste Screenshot Below):

```
root@2cade0d6eaf7:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
```

Step 7 - Check Current User

Command: whoami

Explanation: Displays the currently logged-in user.

Expected Output: returns the current user

Screenshot Placeholder (Paste Screenshot Below):

```
root@2cade0d6eaf7:/# whoami
root
root@2cade0d6eaf7:/#
```

Step 8 - Create File

Command: touch notes.txt

Explanation: Creates an empty file named notes.txt.

Expected Output: file creation through terminal

Screenshot Placeholder (Paste Screenshot Below):

```
root@2cade0d6eaf7:/# touch notes.txt
root@2cade0d6eaf7:/# ls
bin  boot  dev  etc  home  lib  media  mnt  notes.txt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@2cade0d6eaf7:/#
```

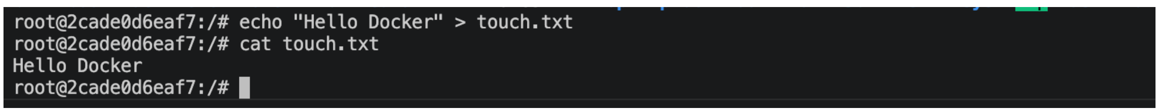
Step 9 - Write Data

Command: echo "Hello Docker" > notes.txt

Explanation: Writes text into a file.

Expected Output: add data to file

Screenshot Placeholder (Paste Screenshot Below):



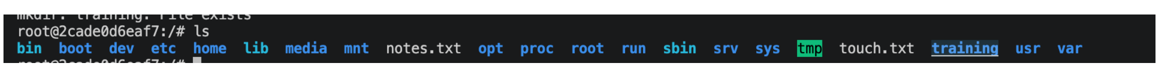
Step 10 - Create Folder

Command: mkdir training

Explanation: Creates a new directory.

Expected Output: Folder creation through terminal

Screenshot Placeholder (Paste Screenshot Below):



Step 11 - View Processes

Command: ps -ef

Explanation: Displays running processes inside the container.

Expected Output: view all process in full format mode

Screenshot Placeholder (Paste Screenshot Below):

```
root@2cade0d6eaf7:/# ps -ef
UID        PID  PPID  C STIME TTY          TIME CMD
root         1     0  0 05:27 pts/0    00:00:00 bash
root        18     1  0 05:34 pts/0    00:00:00 ps -ef
```

Step 12 - Install Software

Commands: apt update && apt install curl -y

Explanation: Updates package repositories and installs curl.

Expected Output: update the lastest version to package list and install the package with (-y), yes to all confirmation prompts

Screenshot Placeholder (Paste Screenshot Below):

```
root@2cade0d6eaf7:/# apt update
Get:1 http://ports.ubuntu.com/ubuntu-ports resolute InRelease [136 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports resolute-updates InRelease [137 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports resolute-backports InRelease [136 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports resolute-security InRelease [137 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports resolute/main arm64 Packages [1860 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports resolute/restricted arm64 Packages [231 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports resolute/universe arm64 Packages [19.9 MB]
Get:8 http://ports.ubuntu.com/ubuntu-ports resolute/multiverse arm64 Packages [298 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports resolute-updates/main arm64 Packages [318 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports resolute-updates/restricted arm64 Packages [328 kB]
Get:11 http://ports.ubuntu.com/ubuntu-ports resolute-updates/universe arm64 Packages [178 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports resolute-updates/multiverse arm64 Packages [3592 B]
Get:13 http://ports.ubuntu.com/ubuntu-ports resolute-security/main arm64 Packages [270 kB]
Get:14 http://ports.ubuntu.com/ubuntu-ports resolute-security/universe arm64 Packages [147 kB]
Get:15 http://ports.ubuntu.com/ubuntu-ports resolute-security/restricted arm64 Packages [324 kB]
Fetched 24.4 MB in 9s (2849 kB/s)
All packages are up to date.
```

```
root@2cade0d6eaf7:/# apt install curl -y
Installing:
  curl

Installing dependencies:
  bash-completion  libcurl4t64       libhogweed6t64  libkrb5-3         libnettle8t64  librtmp1              libssh2-1t64   publicsuffix
  ca-certificates  libffi8           libidn2-0       libkrb5support0   libnghttp2-14  libsasl2-2            libtasn1-6
  krb5-locales     libgnutls30t64    libk5crypto3    libldap-common    libp11-kit0    libsasl2-modules      libunistring5
  libbrotli1       libgssapi-krb5-2  libkeyutils1    libldap2          libpsl5t64     libsasl2-modules-db   openssl

Suggested packages:
  gnutls-bin  krb5-user                    | libsasl2-modules-gssapi-heimdal  libsasl2-modules-otp
  krb5-doc    libsasl2-modules-gssapi-mit  libsasl2-modules-ldap             libsasl2-modules-sql

Summary:
  Upgrading: 0, Installing: 30, Removing: 0, Not Upgrading: 0
  Download size: 6474 kB
  Space needed: 21.1 MB / 448 GB available
```

```
done.
root@2cade0d6eaf7:/# apt list --installed
apt/resolute,now 3.2.0 arm64 [installed]
base-files/resolute-updates,now 14ubuntu6.1 arm64 [installed]
base-passwd/resolute,now 3.6.8 arm64 [installed]
bash-completion/resolute,now 1:2.16.0-8build1 all [installed,automatic]
bash/resolute,now 5.3-2ubuntu1 arm64 [installed]
bsdutils/resolute,now 1:2.41.3-3ubuntu2 arm64 [installed]
ca-certificates/resolute-updates,resolute-security,now 20260601~26.04.1 all [installed,automatic]
coreutils-from-uutils/resolute,now 0.0.0~ubuntu25 all [installed]
coreutils/resolute,now 9.5-1ubuntu2+0.0.0~ubuntu25 all [installed]
curl/resolute-updates,resolute-security,now 8.18.0-1ubuntu2.1 arm64 [installed]
```

Step 13 - Networking

Commands: hostname, hostname -I

Explanation: Displays hostname and IP address.

Expected Output: displays the current host and their IP

Screenshot Placeholder (Paste Screenshot Below):

```
root@2cade0d6eaf7:/# hostname
2cade0d6eaf7
root@2cade0d6eaf7:/# hostname -I
172.17.0.2
root@2cade0d6eaf7:/#
```

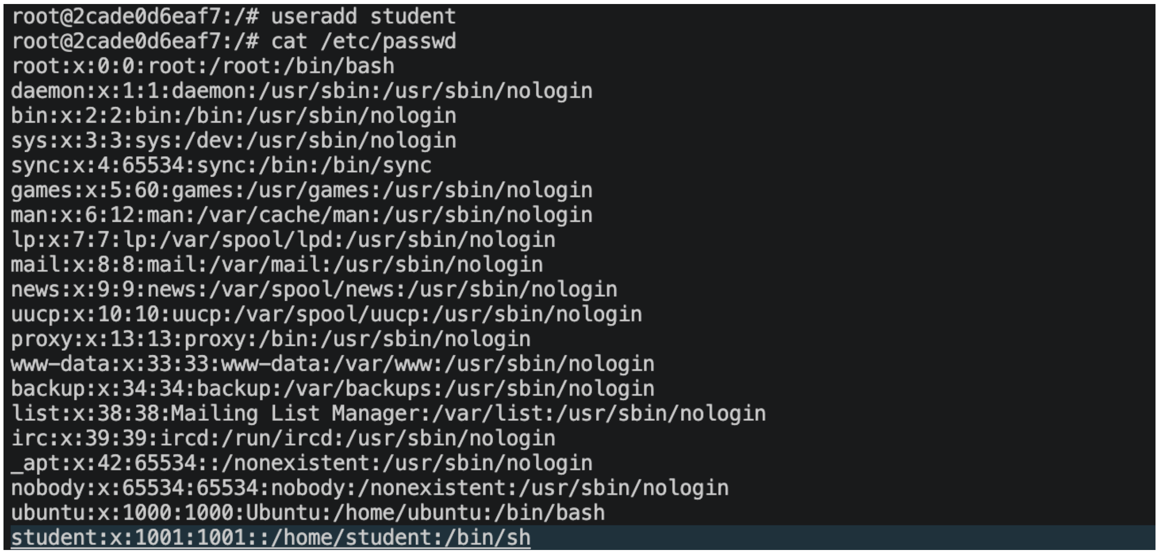
Step 14 - Create User

Command: useradd student

Explanation: Creates a new Linux user.

Expected Output: add student has a linux user

Screenshot Placeholder (Paste Screenshot Below):



```
root@2cade0d6eaf7:/# useradd student
root@2cade0d6eaf7:/# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
student:x:1001:1001::/home/student:/bin/sh
```

Step 15 - Permissions

Command: chmod 777 secret.txt

Explanation: Changes file permissions to full access.

Expected Output: change mode to read (4), write (2), execute (1) = 7 permissions to owner, group and other users

Screenshot Placeholder (Paste Screenshot Below):

```
root@2cade0d6eaf7:/# touch secret.txt
root@2cade0d6eaf7:/# chmod 777 secret.txt
root@2cade0d6eaf7:/# ls -l
total 68
lrwxrwxrwx   1 root root    7 Apr 20 08:46 bin -> usr/bin
drwxr-xr-x   2 root root 4096 Apr 20 08:46 boot
drwxr-xr-x   5 root root  360 Jun 22 05:27 dev
drwxr-xr-x   1 root root 4096 Jun 22 05:38 etc
drwxr-xr-x   3 root root 4096 Jun 10 03:32 home
lrwxrwxrwx   1 root root    7 Apr 20 08:46 lib -> usr/lib
drwxr-xr-x   2 root root 4096 Jun 10 03:31 media
drwxr-xr-x   2 root root 4096 Jun 10 03:31 mnt
-rw-r--r--   1 root root    0 Jun 22 05:32 notes.txt
drwxr-xr-x   2 root root 4096 Jun 10 03:31 opt
dr-xr-xr-x 239 root root    0 Jun 22 05:27 proc
drwx------   2 root root 4096 Jun 10 03:32 root
drwxr-xr-x   4 root root 4096 Jun 10 03:32 run
lrwxrwxrwx   1 root root    8 Apr 20 08:46 sbin -> usr/sbin
-rwxrwxrwx   1 root root    0 Jun 22 05:40 secret.txt
```

Step 16 - Environment Variables

Command: export COMPANY=Training

Explanation: Creates an environment variable.

Expected Output: add environment variable to env common to processor and its child processors

Screenshot Placeholder (Paste Screenshot Below):

```
root@2cade0d6eaf7:/# export COMPANY=Training
root@2cade0d6eaf7:/# echo $COMPANY
Training
root@2cade0d6eaf7:/# env
HOSTNAME=2cade0d6eaf7
COMPANY=Training
```

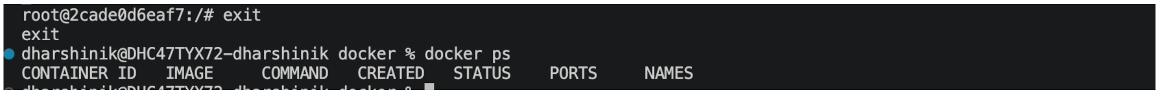
Step 17 - Exit Container

Command: exit

Explanation: Leaves the container shell.

Expected Output: stops the container

Screenshot Placeholder (Paste Screenshot Below):



```
root@2cade0d6eaf7:/# exit
exit
dharshinik@DHC47TYX72-dharshinik docker % docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```